

Shoaib Asghar

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Professional Summary

Computer Science graduate from **Information Technology University (ITU)** with expertise in backend development, full-stack engineering, and AI applications. Backend-focused Python Developer experienced in building scalable REST APIs, AI/RAG pipelines, geospatial systems, and full-stack applications using Django, FastAPI, React, PostgreSQL, and Docker. Seeking Backend, Full-Stack, or AI Engineering roles.

Experience

AI/ML Intern

Neuralige | Mar 2026 – June 2026

🌐 neuralige.ai [in](#) neuralige

Python, FastAPI, Django, React, PostgreSQL, Docker, LangChain, ChromaDB, OpenRouter, ASP.NET

- Completed intensive Python training before contributing to production-level AI and full-stack applications.
- Built scalable REST APIs and backend systems using FastAPI and Django.
- Developed a Hotel Management System using FastAPI, React, PostgreSQL, Docker, and JWT Authentication.
- Built the backend for a Fuel Optimizer, integrating geospatial APIs for route optimization.
- Developed CLARA, a RAG-based backend using LangChain, ChromaDB, and OpenRouter LLMs for intelligent document retrieval.
- Contributed to the Mantra AI Therapist application and the U.S.-based Rank One platform, developing backend modules and collaborating in an Agile environment using Jira

Education

Bachelor of Science in Computer Science

Information Technology University – Lahore, Pakistan

Sept 2022 –June 2026

- Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Database Systems, Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning, Information Security

Projects

Fuel Route Optimizer – Full-Stack Fuel Planning Platform

Tech Stack: Django 5, Django REST Framework, React, PostgreSQL, Docker, OpenRouteService

- Developed a full-stack fuel route planning platform that optimizes fuel stops across multi-stop trips.
- Built REST APIs with Django REST Framework and integrated geospatial routing using OpenRouteService.
- Developed a responsive React frontend with interactive maps, live fuel-cost estimation, and Docker-based deployment.

CLARA – Enterprise AI RAG Assistant

Tech Stack: LangChain, ChromaDB, OpenRouter LLMs, Tavily API, Python

- Developed an enterprise Retrieval-Augmented Generation (RAG) backend for intelligent document retrieval and question answering.
- Integrated LangChain, ChromaDB, OpenRouter LLMs, and Tavily Search to generate accurate, context-aware responses.
- Designed a scalable retrieval pipeline supporting both internal knowledge and live web search.

Smart Car Assistant (Final Year Project)

Tech Stack: Python, OpenCV, YOLOv8, Raspberry Pi

- Developed a real-time driver assistance system using YOLOv8 for object and lane detection on Raspberry Pi.
- Optimized the computer vision pipeline for embedded hardware and real-time performance.
- Selected by the Head of the Computer Science Department for departmental investment based on technical innovation and real-world impact.

Technical Skills

Languages: Python, SQL, JavaScript, C/C++, Bash

Backend: Django, Django REST Framework, FastAPI, Flask, Node.js, Express.js

Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, MongoDB, ChromaDB, SQL Server

Data & AI/ML: Pandas, NumPy, Scikit-learn, LangChain, YOLOv8, OpenCV

DevOps & Tools: Docker, Docker Compose, Git, Postman, JWT, Alembic

APIs: REST APIs, OpenRouteService, Geopy, Tavily API

Achievements

- Web Development Certification – Udemy
- Ranked #1 Intern Project – CLARA, Neuralige
- Final Year Project Selected for Departmental Investment by the Head of Department, CS, ITU Punjab